Lab 04 - Prioritise Problems by Severity, Frequency and Impact

Course: DevOps Foundation Training (TGS-2024049184) Version: v4.0 Topic: 04 Estimated time: 35-45 minutes

Workplace scenario

Northstar Retail can fund only two problem-remediation initiatives this quarter. Leaders disagree because each team uses a different definition of priority.

Objective

Apply K3 and A3 by defining evidence, then K4 and A4 by applying a transparent prioritisation rule.

What you will produce

A ranked problem_priority.csv and priority_matrix.png with documented scoring assumptions.

Prerequisites

- R 4.x available as Rscript.
- Work from this lab folder so relative paths resolve.
- Use only the supplied synthetic data; no internet connection or external package is required.
- Preserve the source data. Make experimental changes in a copied file, then restore it.

Folder map

```
Lab-04-problem-prioritisation/
■■■ README.md / README.pdf
■■■ CHECKLIST.md / CHECKLIST.pdf
■■■ problem_prioritisation.R
■■■ data/
■■■ output/            # reproducible evidence produced by the R script
```

Detailed procedure

Step 1 - Review problems.csv and confirm all scoring fields use the documented 1-5 scale.

```
Rscript -e 'd <- read.csv("data/problems.csv"); summary(d)'
```

Record what you observed before continuing.

Step 2 - Run problem_prioritisation.R to calculate impact-weighted priority and rank the backlog.

```
Rscript problem_prioritisation.R
```

Record what you observed before continuing.

Step 3 - Inspect the top two problems and explain why they rank above a severe but low-confidence item.

```
Rscript -e 'print(read.csv("output/problem_priority.csv"))'
```

Record what you observed before continuing.

Step 4 - Open the severity-frequency matrix and compare visual position with the final score.

```
open output/priority_matrix.png
```

Record what you observed before continuing.

Step 5 - Run one sensitivity test by increasing an effort estimate or reducing confidence, then restore the source file.

```
Rscript problem_prioritisation.R
```

Record what you observed before continuing.

Step 6 - Document the scoring policy, selected initiatives and trade-off in CHECKLIST.md.

```
open CHECKLIST.md
```

Record what you observed before continuing.

Evidence to retain

[ ] output/problem_priority.csv

[ ] output/priority_matrix.png

[ ] completed CHECKLIST.md

Acceptance criteria

- Formula is explicit
- Unique rank for all problems
- Top two selected with evidence
- Sensitivity test is explained

Verification

Every score is reproducible from the published formula; ranks are unique; the top two recommendations cite impact, evidence confidence and effort.

Troubleshooting

- cannot open file: confirm your terminal is in this lab folder.
- Rscript: command not found: install R, reopen the terminal and run Rscript --version.

- A validation stop(...) message is evidence that the input broke a stated rule; inspect the named field rather than bypassing the check.
- If an output looks wrong, compare row counts and IDs with the source CSV before changing the analysis.

Clean rerun

Delete only the files inside output/, then run the R script again. The source CSV and script remain unchanged.